MILITARY SPECIFICATION

CLIP, ELECTRICAL, GRID AND ANODE; GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for grid and anode types of electrical clips for electron tubes

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

QQ-C-533 — Cooper-Beryllium Alloy Strip

QQ-P-416 — Plating Cadmium (Electrodeposited).

UU-B-36 — Bags, Papers Grocers.

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-636 — Box, Fiberboard

PPP-B-676 — Boxes, Set-Up, Paperboard.

PPP-T-60 — Tape, Pressure Sensitive Adhesive, Waterproof — for Packaging and Sealing

MILITARY

MIL-I-10 — Insulating Materials, Electrical, Ceramic, Class L.

MIL-P-116 — Preservation, Methods of.

MIL-R-5847 — Rubber, Silicone; Low- and High-Temperature and Tear Resistant

MIL-T-10727 — Tin Plating, Electrodeposited or Hot-dipped, for Ferrous and Nonferrous Metals

MIL-W-16878 (Navy) — Wire, Electrical, Insulated, High Temperature.

MIL-F-22191 — Films, Transparent, Flexible, Heat Sealable, for Packaging Applcations.

MIL-C-45662 — Calibration System Requirements.

FSC 5940

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspcetion by Attributes.

MIL-STD-129 — Marking for Shipment and Storage

MIL-STD-130 — Identification Marking of U.S Military Property

MIL-STD-202 — Tset Methods for Electronic and Electrical Component Parts

MIL-STD-456 — Electronic Parts, Date and Source Coding for.

MIL-STD-810 — Environmental Test for Aerospace and Ground Equipment.

MS33586 — Metals, Definition of Dissimilar.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets.

3.2 Material. The material for each part shall be as specified (see 3.1). When a definite material is not specified, a material which will enable the clips to meet the requirements of this specification shall be used Acceptance or approval of a constituent material shall not be construed as a guarantee of the acceptance of the finished product. Materials shall be non-nutrient to fungus growth. Submission of certification that materials used are non-nutrient to fungus growth will be considered compliance with this requirement without testing. If certification is not furnished, clips shall be subjected to the fungus-resistance test of method 508, procedure I, of MIL-STD-810

3.2.1 *Ceramic* Ceramic material shall conform to MIL-I-10, grade 1422 or better.

3.2.2 *Metals*

3.2.2.1 *Beryllium-copper alloy* Beryllium-copper alloy shall conform to QQ-C-533 and shall be suitably heat treated after forming and prior to cleaning and plating.

3.2.3 *Silicone rubber*. Silicone rubber shall conform to class III, of MIL-R-5847.

3.2.4 *Corrosion-resistance* The clips shall be fabricated of a corrosion-resistant material or be treated to prevent corrosion formation

3.3. Design and construction. The clips shall be of the material, design, construction, and physical dimensions specified (see 3.1)

3.4 Finish. Finish shall be electro-tin plate as specified in MIL-T-10727, type I. The clips shall be plated so that before and after the environmental tests specified herein, there shall be evidence of peeling, cracking or corrosion. No additional insulating material shall be applied unless specifically approved by the procuring activity

3.4.1 *Cadmium plating*. Cadmium plating shall conform to class 2, type II of QQ-P-416.

3.5 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided The use of dissimilar metals which, in contact, tend toward

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.

active electrolytic corrosion (particularly brass, copper, or steel used in metals) shall be as specified in MS33586.

3.6 Salt spray (corrosion). When clips are tested as specified in 4.6.2.1, the base metal shall show no evidence of corrosion or blistering of plated surfaces.

3.7 Vibration. When tested as specified in 4.6.2.2 the clip shall display no evidence of cracking, breaking, loosening of parts, wear or damage to the clip, nor loss of electrical continuity of the contact circuit for a period greater than 10 microseconds

3.8 Thermal shock. When tested as specified in 4.6.2.3, the clip shall retain a force on the test plug of not less than 80 percent as measured before the test There shall be no cracking or chipping of the ceramic insulated clips

3.9 Shock When tested as specified in 4.6.2.4, there shall be no evidence of mechanical damage to the clip The clip shall retain a force on the test plug of not less than 80 percent as measured before the test

3.10 Retaining and release force. When clips are tested as specified in 4.6.2.5, the force required to couple and uncouple the clip shall be within the limits specified (see 3.1).

3.11 Dielectric withstanding voltage. When clips are tested as specified in 4.6.2.6, there shall be no evidence of damage, arcing, or breakdown.

3.12 Identification marking. The clips shall be legibly and durably marked with the part number and the manufacturer's name or trademark in accordance with MIL-STD-130 and with the code symbol in accordance with MIL-STD-456

3.13 Workmanship. The clips shall show no evidence of cracks, fissures, peeling, or chipping of the plating or finish. When examined (see 4.6.1), there shall be no evidence or poor molding, poor fabrication techniques, or foreign material

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements

4.1.1 *Test equipment and inspection facilities.* Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required acceptance inspection The supplier shall establish adequate calibration of inspection equipment to the satisfaction of the Government Calibration of the standard which control the accuracy of the inspection equipment shall comply with the requirements of MIL-C-45662.

4.2 Classification. The examination and test of electrical clips shall be as follows.

(a) Component-materials inspection (see 4.3)

(b) Quality conformance inspection (see 4.5)

1. Inspection of products for delivery (see 4.5.1)

2. Inspection of preparation for delivery (see 4.5.2).

4.3 Component-materials inspection. Com-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.

ponent-materials inspection shall consist of verification that the component materials listed in table I, used in fabricating the electrical clips, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE I. *Component-materials inspection*

Component material	Requirement paragraph	Applicable specification
Insulation:		
Ceramic	3.2.1	MIL-I-10
Silicone rubber	3.2.3	MIL-R-5847
Metals		
Beryllium-copper alloy	3.2.2.1	QQ-C-533
Finish		
Tin plating	3.4	MIL-T-10727

4.4. **Test conditions.** Test conditions shall be as specified in MIL-STD-202 unless otherwise specified herein.

4.5 **Quality conformance inspection.**

4.5.1 *Inspection of product for delivery.* Inspection of product for delivery shall consist of groups A and B. Sampling plans and procedures for inspection shall be as specified in MIL-STD-105.

4.5.1.1 *Inspection lot.* An inspection lot shall consist of all the clips of the same material, configuration and dimensions, manufactured under essentially the same conditions and submitted for quality conformance inspection and testing at one time.

4.5.1.2 *Rejected lots.* If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units, as applicable, and reinspect. Such lots shall be separated from new lots and shall be clearly identified as reinspected lots. Rejected lots shall be inspected using tightened inspection.

4.5.1.3 *Group A inspection.* Group A inspection shall consist of an examination of product (see table II). The inspection level shall be II (see 3.1, 3.12, and 3.13).

TABLE II. *Group A inspection*

Examination or test	Requirement paragraph	Method paragraph
Examination of product	3.1, 3.12, and 3.13	4.6.1

4.5.1.3.1 *Sampling plan.* Statistical sampling and inspection shall be in accordance with MIL-STD-105 for normal inspection. The acceptable quality levels (AQL) shall be 1.0 and 4.0 percent defective for major and minor defects respectively. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.3.2 *Disposition of sample units.* Sample units which have been subjected to group A inspection shall be delivered on the contract or order.

4.5.1.4 *Group B inspection.* Group B inspection shall consist of the tests specified in table III in the order shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.

TABLE III *Group B inspection*

Test	Requirement paragraph	Method paragraph
Group I		
Vibration	3 7	4.6.2.2
Thermal shock	3 8	4 6.2 3
Shock	3 9	4 6.2.4
Retaining and release force	3 10	4 6 2.5
Group II		
Dielectric withstanding voltage	3 11	4 6 2 6
Salt spray	3 6	4.6 2.1

4.5.1.4.1 *Sampling plan* Each year six sample units shall be selected from the first lot and divided into two subgroups of three each. No failures shall be allowed Group B inspection shall be performed on sample units that have passed group A inspection, unless the Government considers it more practical to select a separate sample from the lot for group B inspection

4.5.1.4.2 *Disposition of sample units* Sample units which have been subjected to group B inspection shall not be delivered on the contract or order

4.5.2 *Inspection of preparation for delivery* Sample items and packs shall be selected in accordance with MIL-P-116 to verify conformance to section 5 of this specification

4.6 Methods of examination and test.

4.6.1 *Examination of product.* Clips shall be examined for compliance with the applicable design, construction, physical dimensions, marking and workmanship requirements (see 3.1, 3.12, and 3 13)

4.6.2 *Test procedures.*

4.6.2.1 *Salt spray (corrosion) (see 3 6)* Sample clips shall be tested in accordance with MIL-STD-202, method 101, condition A. Salt concentration shall be 5 percent. After the test the clips shall be examined for exposure of base metal or blistering of the plated surface

4.6.2.2 *Vibration, high frequency (see 3.7).* Clips shall be tested in accordance with MIL-STD-202, method 204 and with the following exceptions and details

(a) Specimens shall be mounted in a normal manner to a test plug which shall be fastened to the vibration machine The clip shall be mated to the plug or tube without the use of safety wire or other supplementary locking devices The lead wire shall be supported by a stationary frame not closer than 12 inches from the mated clip and tube cap or test plug The wire shall conform to MIL-W-16878

(b) Test condition letter D

(c) The clips shall be wired and current allowed to flow at the maximum operating rated voltage and current of the tube with which the clip mates

(d) The vibration shall be conducted in each of three mutually perpendicular directions at the following temperature conditions

(1) —65° C — 25 percent of the time.

(2) −265 C — 25 per cent of the time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.

(3) normal temperature conditions — 50 percent of the time.

(e) The circuit shall be monitored for circuit continuity before, during, and after the vibration. A discontinuity of contact or interruption of current flow for more than ten microseconds shall not be allowed.

4.6.2.3 *Thermal shock.* Clips shall be tested in accordance with MIL-STD-202, method 107, test condition C and shall meet the requirements of 3.8.

4.6.2.4 *Shock.* Clips shall be tested in accordance with method 202 of MIL-STD-202 The following details and exceptions shall apply·

(a) *Mounting of specimens* By their normal mounting means with the test plug fastened to the shock machine and the clip coupled to the test plug

(b) *Acceleration* 50 gravity units.

(c) *Number of blows* Four

(d) *Direction of blows.* One direction shall be in the axis of the test plug, away from the open end of the clip, and the other direction shall be perpendicular to the axis of the test plug.

(e) *Measurements before and after shock.* Clips shall be removed from the test plug parts and the withdrawal force recorded.

4.6.2.5 *Retaining and release force for electrical clips* The clips shall be rigidly mounted so as to permit coupling and uncoupling of the test plugs. The force to couple and uncouple the clips on the maximum size test plug shall be within the limits specified (see 3.1). The clip shall be coupled and uncoupled 500 times. The maximum size test plug shall have a diameter of no less than the tube cap diameter +.015 inches. A minimum size test plug shall be inserted, and the gradually applied release force shall be measured. This force shall be as specified by the retaining and release force of the specification sheets. The minimum size test plug shall have a diameter of no more than the tube cap diameter, —.007 inches

4.6.2.6 *Dielectric withstanding voltage.* Clips shall be tested in accordance with method 301 of MIL-STD-202 The following details and exceptions shall apply

(a) Special preparation or conditions —

(1) The maximum relative humidity shall be 50 percent

(2) The center of contact of the clips shall be positioned in such a manner as to simulate actual assembly conditions.

(3) Precautions shall be taken to prevent air gap voltage breakdown.

(b) Magnitude of test voltage — 3,500 volts (RMS) at sea level, 1,500 volts (RMS) at 50,000 feet, 750 volts (RMS) at 70,000 feet. Voltage shall be applied instantaneously

(c) Nature of potential — Alternating current.

(d) Points of application of test voltage — As specified on figure in specification sheets.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6 1).

5.1.1 *Level A.*

5.1.1.1 *Cleaning* Clips shall be cleaned in accordance with MIL-P-116, process C-1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.

5.1.1.2 *Drying.* The procedure selected shall not be injurious to the item

5.1.1.3 *Preservative application* None required.

5.1.1.4 *Unit packaging* Clips shall be individually packaged in accordance with MIL-P-116, method III The unit package may be accomplished by the use of paperboard boxes, PPP-B-566 or PPP-B-676, paper envelopes, UU-B-36, plastic envelopes, MIL-F-22191 or skin or blister packaging in such a manner that the packages may be separated for individual issue

5.1.2 *Level C* Clips shall be preserved and packaged in a manner that will afford adequate protection against corrosion deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use This package may conform to the supplier's commercial practice for retail distribution when such meets the requirements of this level

5.2 **Packing** Packing shall be level A, B, or C, as specified (see 6 1).

5 2 1 *Level A* The packaged items shall be packed in fiberboard containers conforming to PPP-B-636, weather resistant style optional, special requirement In lieu of the closure and waterproof requirements, closures, and waterproofing shall be accomplished by sealing the center seams ends edges and manufacturer's joints with waterproof tape, 3 inches wide, conforming to PPP-T-60. Banding (reinforcement requirements), excluding metal shall be applied in accordance with the appendix to PPP-B-636

5.2.2 *Level B.* The packaged item shall be packed in fiberboard containers conforming to PPP-B-636 class domestic optional requirements Box closures shall be in accordance with method 1 of the appendix to PPP-B-636.

5.2.3 *Level C* The packaged item shall be packed in a manner that will afford adequate protection against damage during direct shipment form the supply source to the first receiving activity for immediate use. This pack shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such conforms to the requirements of this level

5.2 4 *General* Exterior containers shall be uniform in size and shape shall be of a minimum cube and tare consistent with the protection required and shall contain equal quantities of identical items to the greatest extent practical

5.3 **Marking (see 6.1)** In addition to any special marking required by the contract or order unit packages intermediate and exterior containers shall be marked in accordance with MIL-STD-129

5.4 **Workmanship.** The packaged items shall be prepared for delivery in a manner that is uniform in quality and free from defects that will affect service, life or appearance of the items, packapes or packs.

6. NOTES

6.1 **Ordering data.** Procurement document shall specify the following

(a) Tittle, number and date of this specification

(b) Title, number, and date of the applicable military specification sheet and the complete part number (see 3.1).

(c) Levels of preservation, packing, and packaging and applicable marking (see section 5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.

Custodians:

Army—EL

Navy—SH

Air Force—85

Review activities:

Army

Navy

Air Force—17

User activities:

Army

Navy

Air Force

Preparing activity:

Army—EL

Project No. 5940-0272

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.

FOLD

HEADQUARTERS
U S ARMY ELECTRONICS MATERIEL SUPPORT AGENCY
FORT MONMOUTH, NEW JERSEY 07703

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding Officer
U. S. Army Electronics Materiel Support Agency
Fort Monmouth, New Jersey 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

SUBMITTED BY (*Printed or typed name and activity*) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:21Z
Check the source to verify that this is the current version before use.